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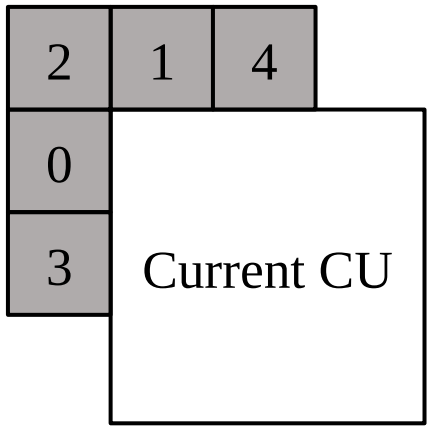
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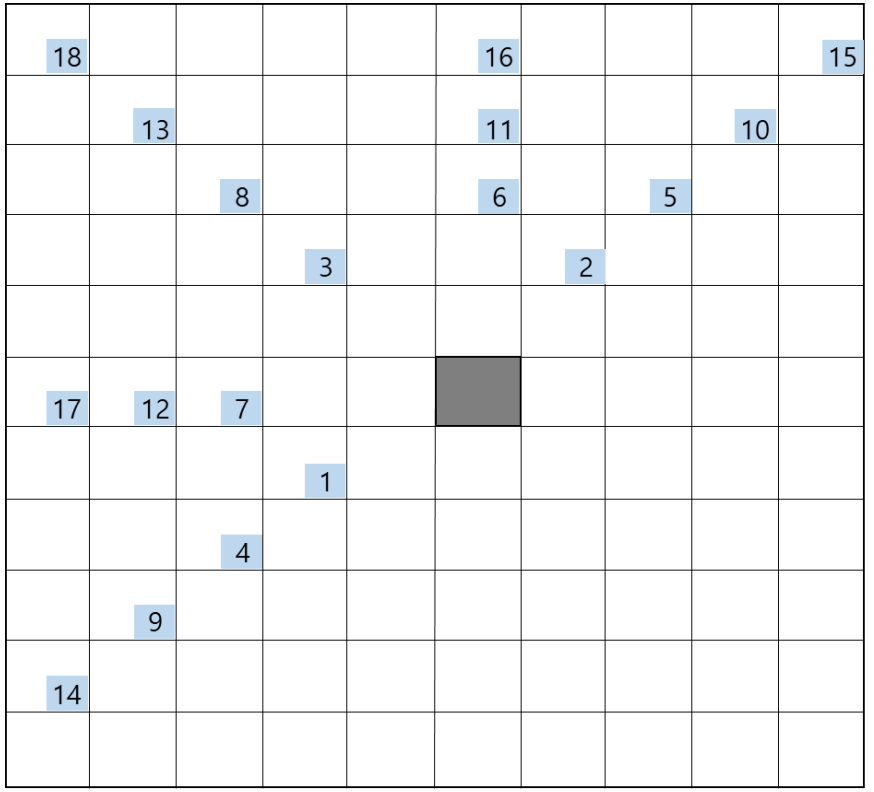
AhG12: The Modification of BV Candidate List Construction

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- **In ECM-13.0, BV candidate list is constructed in DIMD, OBIC, SGPM, and IntraTMP tools.**
  - The BV candidate list is composed of BVs derived from conventional adjacent and non-adjacent spatial candidate blocks.
    - ✓ Conventional adjacent spatial candidate blocks: 5 blocks
    - ✓ Non-adjacent spatial candidate blocks: 18 blocks





<Conventional adjacent spatial candidate blocks>

<Non-adjacent spatial candidate blocks>

Figure 1. Spatial candidate blocks used in BV candidate list construction

- Two methods to modify the adjacent spatial candidate blocks in the process of BV candidate list construction.
  - **Method 1)** Replaced conventional adjacent spatial candidate blocks with the adjacent spatial candidate blocks **used in OBIC mode.**
  - **Method 2)** Modified the adjacent spatial candidate blocks of Method 1 **without exceptional cases.**

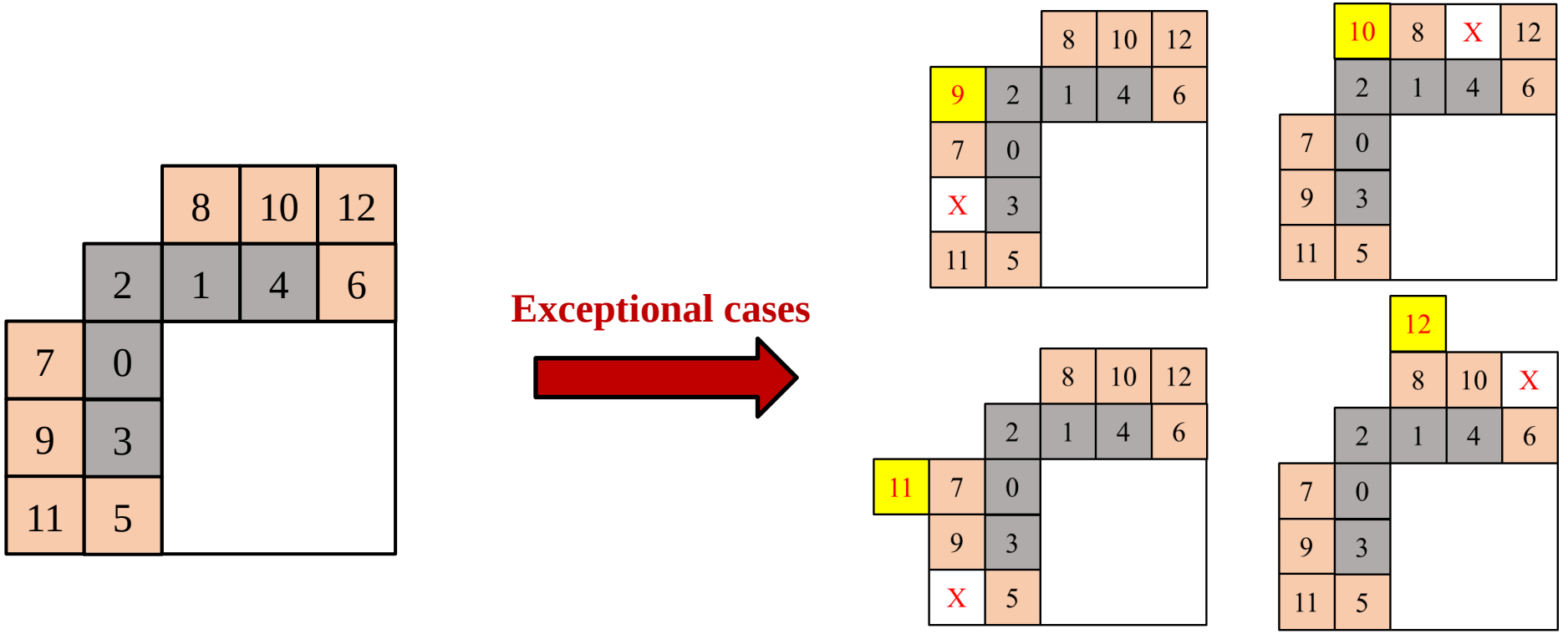


Figure 2. Adjacent spatial candidate blocks used in OBIC mode

- **Performances on top of ECM-13.0**

Method 1)

	All Intra Main 10						
	Over ECM-13.0						
	Y	U	V	EncT	DecT	EncVmPeak	DecVmPeak
Class A1	-0.01%	0.00%	0.03%	100.2%	101.8%	101.3%	100.1%
Class A2	-0.01%	0.01%	-0.02%	100.4%	101.7%	98.4%	100.1%
Class B	-0.01%	-0.06%	-0.14%	100.5%	101.6%	103.0%	100.0%
Class C	-0.01%	0.00%	0.03%	100.6%	101.4%	100.0%	100.0%
Class E	-0.03%	0.08%	-0.15%	100.6%	101.5%	101.0%	100.0%
Overall	-0.02%	0.00%	-0.06%	100.5%	101.6%	100.9%	100.0%
Class D	0.01%	0.06%	-0.03%	100.9%	101.7%	100.0%	100.0%
Class F	-0.07%	0.03%	0.01%	100.8%	101.4%	99.5%	100.0%
Class TGM	**-0.10%**	-0.07%	-0.07%	100.4%	101.6%	101.3%	100.2%

Method 2)

	All Intra Main 10						
	Over ECM-13.0						
	Y	U	V	EncT	DecT	EncVmPeak	DecVmPeak
Class A1	0.00%	-0.01%	-0.04%	100.3%	100.3%	99.7%	99.9%
Class A2	-0.01%	-0.06%	-0.01%	100.0%	100.4%	98.9%	100.0%
Class B	0.00%	0.00%	0.01%	100.3%	100.6%	103.0%	100.0%
Class C	0.00%	0.02%	0.09%	100.1%	100.4%	99.5%	100.0%
Class E	-0.03%	-0.02%	0.00%	100.2%	100.6%	103.2%	99.9%
Overall	-0.01%	-0.01%	0.01%	100.2%	100.5%	101.0%	100.0%
Class D	0.01%	0.06%	-0.19%	100.4%	100.3%	100.1%	100.0%
Class F	**-0.11%**	0.05%	0.00%	100.6%	100.4%	99.5%	99.9%
Class TGM	**-0.08%**	-0.11%	-0.10%	100.5%	100.5%	101.2%	100.0%

- **Summary**
  - The use of the modified adjacent spatial candidate block in the BV candidate list was presented in this contribution.
  - Overall performances on ECM-13.0 are reported as below:
    - ✓ Method (1) AI: {-0.02%, 0.00%, -0.06%, 100.5% EncT, 101.6% DecT} for Overall
      {-0.07%, 0.03%, 0.01%, 100.8% EncT, 101.4% DecT} for class F
      {**-0.10%**, -0.07%, -0.07%, 100.4% EncT, 101.6% DecT} for class TGM
    - ✓ Method (2) AI: {-0.01%, -0.01%, 0.01%, 100.2% EncT, 100.5% DecT} for Overall
      {-0.11%, 0.05%, 0.00%, 100.6% EncT, 100.4% DecT} for class F
      {-0.08%, -0.11%, -0.10%, 100.5% EncT, 100.5% DecT} for class TGM

- **It is suggested that the proposed method (2) can be investigated in the next round of EE2.**

- **Thanks to LGE for cross-checking!**

THANK YOU!

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